



OptiEDT

ACADEMIC SCHEDULING

Interface guide

How to use the system, screen by screen

A short walkthrough of every screen. No background in optimisation, programming or artificial intelligence is needed to read it.

Project OptiEDT – timetable generation for a Tunisian public faculty (LMD)
Author Mohamed Jawad Touri
Version Interface V2, frozen 13 August 2026
Source docs/supervisor/INTERFACE_WALKTHROUGH.md

What OptiEDT is

OptiEDT builds weekly timetables for a university department.

You give it the department's data — courses, teachers, rooms, groups and the hours in a week. It produces several complete timetables that break none of the department's rules, puts them in order from best to worst against quality criteria you can inspect, and explains in plain language why one is better than another.

Three things are worth knowing up front:

- **Every timetable it produces is valid.** No teacher is in two rooms at once, no group has two sessions at the same time, no room is double-booked. This is guaranteed, not checked afterwards.
- **It produces several, not one.** They differ in quality, not in validity, and you choose between them.
- **Every number it shows can be checked by hand.** Nothing about the ranking is hidden.

Signing in, and the four roles

Open the address of the application. Four buttons — Timetable Officer, Administrator, Teacher, Student — fill in the matching username for you. **They only fill in a name.** What you are allowed to do comes from the account you actually sign in with.

SCREENSHOT NOT CAPTURED

The sign-in page: product summary on the left, the four role buttons and the username and password fields on the right.

Capture at 1440×900 · route / · signed out

Role	What this person does
Timetable Officer	Loads the department's data, generates timetables, compares them, publishes the chosen one. The main role
Administrator	Manages the academic calendar — holidays, closed half-days — and the user accounts
Teacher	States when they are available to teach, and reads the timetables
Student	Reads the published timetable of their own group

Each role sees only its own screens. A teacher does not see the Generate screen at all.

Finding your way around

The strip down the left is the navigation, in three groups:

- **Timetabling** — producing and reading timetables
- **Inputs** — the information timetables are built from
- **Institution** — calendar and accounts

The white bar across the top always tells you **which screen you are on**, a line about what it is for, and **the one main button for that screen**. That button stays visible while you scroll.

SCREENSHOT NOT CAPTURED

The full application window: left navigation rail, the page bar with its title and main button, and the content below.

Capture at 1440×900 · route /generate · signed in as Timetable Officer

Department data and teacher availability

Department data — Inputs → Department Data

Shows what the system is working from: how many sessions, groups, teachers, courses, rooms, time slots and holidays. To use different data, upload the eleven expected files.

The system **checks every line before recording anything**. If something is wrong it records nothing, keeps the previous data, and lists the exact lines at fault — the file, the line number, the column and the value. An upload can be withdrawn at any time.

Teacher availability — Inputs → Availability

A weekly grid. Green means available to teach; grey means the teacher has said they are not. Click a cell — or drag across several — then press **Save availability**.

Hatched cells are closed by the academic calendar. Nothing is ever scheduled in them, so they cannot be changed here. A teacher sees only their own grid; the Timetable Officer can view any.

SCREENSHOT NOT CAPTURED

The availability grid, with several cells declared unavailable and Saturday afternoon shown as closed.

Capture at 1440×900 · route /availability

Generating a timetable

Timetabling → **Generate**. Choose a search effort — Quick, Standard or Thorough — and press **Generate timetable**. The screen shows three stages as they happen:

1. **Data checks** — a fast check that a timetable is possible at all
2. **Optimisation** — the actual work of placing every session
3. **Score & rank** — measuring the results and putting them in order

This takes a few minutes. You can leave the screen and come back; the run continues. A longer search improves the *quality* of the result — it never affects whether the compulsory rules are respected. Those hold in every result, always.

If a second run says **Queued**, it is waiting for the first to finish. Only one runs at a time, because each uses the whole machine.

SCREENSHOT NOT CAPTURED

The Generate screen after a completed run: the three stages, the run details, and the ranked candidates.

Capture at 1440×900 · route /generate · a completed run

Reading the candidates

Each row of the ranked list shows:

- its **position** — 1 is best;
- the **approach** used: Balanced, Student-friendly or Teacher-friendly;
- a small **bar chart** — seven bars, one per quality criterion, taller is better;
- its **score out of 100**.

The bar chart is there for a reason: two timetables can score almost the same and still be very different. The bars show that at a glance. **Show terms** opens the seven criteria in full; **Publish** adopts one.

Comparing two candidates

Timetabling → Compare. Pick two timetables. The top of the screen puts them side by side with their scores and the difference between them.

Below is the heart of the product: **where the difference comes from**. One row per criterion, with a bar to the right if the first timetable is better on it and to the left if the second is.

At the bottom, two figures: the **sum of the contributions** and the **difference in score**. They are always identical, and a green line says so. That is what makes the ranking defensible — the explanation is the calculation itself, not a summary of it.

Dominance tells you whether one timetable is simply better on every criterion, or whether the ranking is settling a genuine trade-off. **Improve this timetable** lets you change one thing and run again — it never edits the timetable in front of you, so the guarantees still hold.

SCREENSHOT NOT CAPTURED

The Compare screen: the two scores and their difference, the criterion-by-criterion breakdown with its bars, and the AI panel on the right.

Capture at 1440×900 · route /compare · two candidates selected

The AI explanation

On the right of the Compare screen is a panel headed **OptiEDT AI**. It writes explanations in ordinary language: why a candidate scored as it did, what trade-offs it makes, where it is weakest. There are suggested questions, and you can type your own.

Two things matter here, and the screen says both:

- OptiEDT itself does all the calculation — the timetable, the scores, the ranking, the recommendation. The AI **only explains** results that already exist. It decides nothing.
- **It cannot invent a number.** Every figure it writes is checked against the figures the system calculated. If it produces one that does not match, the whole answer is thrown away and you are shown the system's own figures instead, with the reason.

Above every answer is a line saying who wrote it — *Written by the language service* or *Computed by the application*. Read that line first. If the AI is switched off, everything still works: you see the system's own figures, complete, just not written as prose.

Timetables

Timetabling → **Timetables**. The week as a grid, with four views on the tabs:

View	Shows
By teacher	One teacher's week
By group	One group's week
By room	One room's week
Room occupancy	How heavily each room is used

Colours mark the kind of session, and each cell also prints the code, so colour is never the only clue:

 **CM** — a lecture, whole year group  **TD** — a tutorial, one small group  **TP** — practical work, in a laboratory

Print and **Export CSV** are in the top bar. Both carry which run and which candidate they came from, so a printed sheet can always be traced back.

SCREENSHOT NOT CAPTURED

The weekly grid for one group, with lectures, tutorials and practicals visible and the closed half-day hatched.

Capture at 1440×900 · route /timetables · view "By group"

Examinations, and publishing

Examinations — Timetabling → Examinations

Builds the examination calendar: one examination per course, placed inside the examination period set in Administration. You supply nothing here — the candidates are the students of the year group concerned and the supervisor is the teacher who gives the lecture, both taken from the department's data. Choose an effort, press **Generate calendar**; it takes about half a minute.

Publishing

Publishing is the moment the department says *this is the timetable people follow*. Press **Publish** on a candidate from the Generate screen. It appears under **Published** with the full record of what produced it.

Nothing reaches students until it is published.

What a teacher and a student see

A teacher sees three entries: their availability, the timetables, and the comparison. They can read a comparison and its explanation in full, but cannot start a new run — that belongs to the Timetable Officer.

A student sees one screen: **My Timetable** — their own group's published week, which they can print or export. They cannot see draft timetables or other groups. Until the department publishes, the screen says so plainly.

SCREENSHOT NOT CAPTURED

The student's screen: one navigation entry and their own group's published week.

Capture at 1440×900 · route /my-timetable · signed in as Student

In short

1. The Timetable Officer loads the data; teachers state their availability.
2. The officer generates — the system produces several valid timetables and ranks them.
3. The officer compares two, reads why they differ, and asks the AI if a plain-language explanation helps.
4. The officer publishes one.
5. Teachers and students read it, print it, or export it.

On the screenshots. The nine placeholders in this guide are marked rather than filled because the environment that produced this document had no way to capture the running interface. Each names the screen, the route and the window size to capture, so the images can be taken and dropped in without re-writing the text. **No screenshot has been simulated, illustrated or reconstructed** — a picture of a product that does not exactly match the product is worse than a gap that says so.

Source of truth: docs/supervisor/INTERFACE_WALKTHROUGH.md. Regenerate this PDF from docs/supervisor/guide.html.